The requirement to comply with changes implemented in this Revision must be completed by 6 months from the date of this document.

INCH-POUND

MS27750K
31 August 2026
SUPERSEDING
MS27750J
20 June 2016

MILITARY SPECIFICATION SHEET

# RELAYS, ELECTROMAGNETIC, 50 AMPERES, 3 PDT-NO., CENTER OFF, HERMETICALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

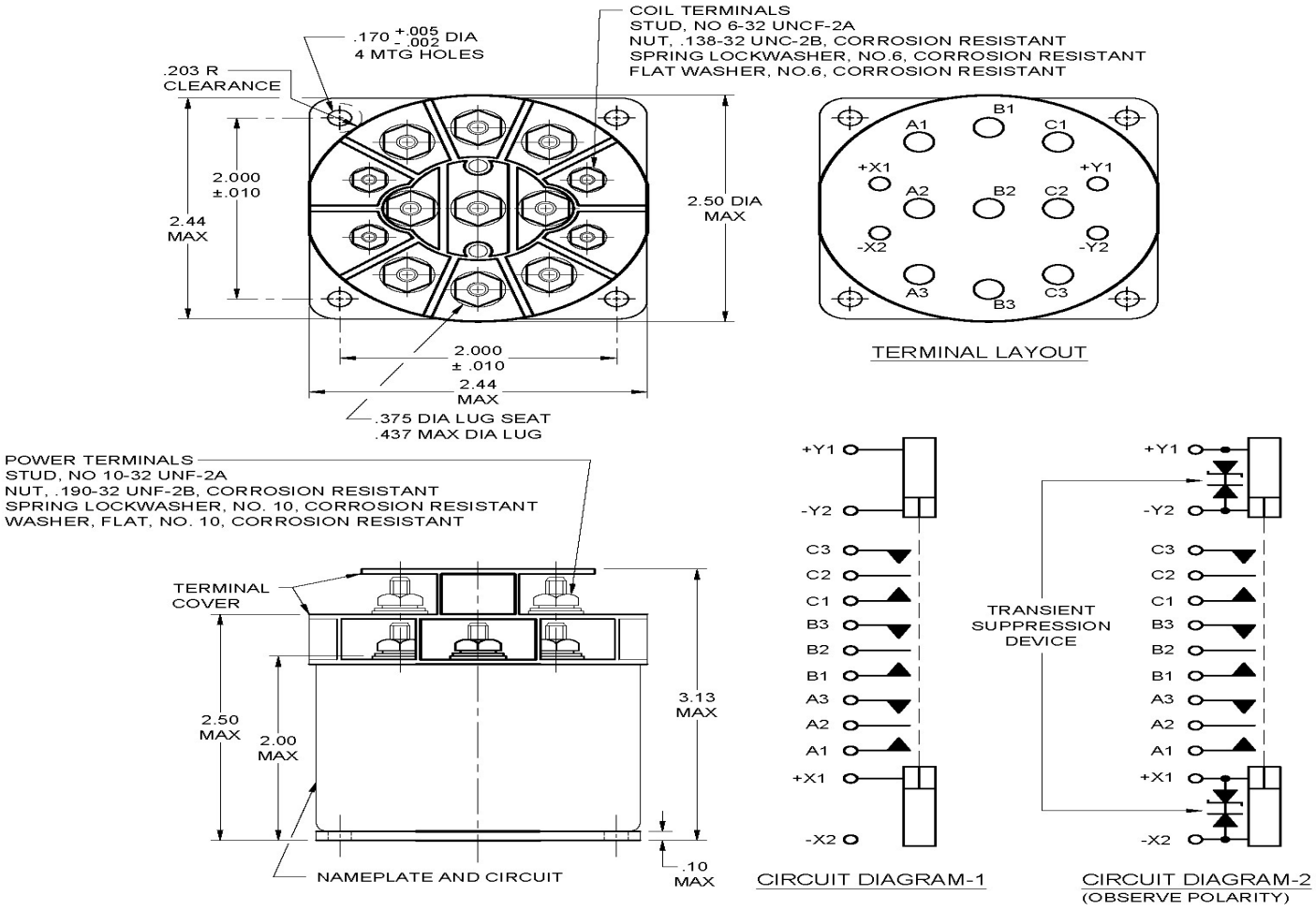



FIGURE 1. Dimensions and configuration.

AMSC N/A

FSC 5945

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| Inches | mm |
|---|---|
| .002 | 0.05 |
| .005 | 0.13 |
| .010 | 0.25 |
| .10 | 2.5 |
| .170 | 4.32 |
| .203 | 5.16 |
| .375 | 9.52 |
| .437 | 11.10 |
| 2.000 | 50.80 |
| 2.44 | 62.0 |
| 2.50 | 63.5 |
| 3.13 | 79.5 |

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
3. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.
4. Referenced Government documents of the issue listed in that issue of the Acquisition Streamlining and Standardization Information System (ASSIST) specified in the solicitation forms a part of this standard to the extent specified herein.
5. Weight includes terminal barriers.
6. Suppression level - the maximum induced transient voltage (back EMF) shall be 42 volts.
7. Metric equivalents are given for general information only.

| MS part number | Type | Coil | Terminal type | Mounting | Max weight in pounds |
|---|---|---|---|---|---|
| MS27750-1 | I | dc | Stud | Plate | .969 |
| MS27750-2 | I | dc | Stud | Plate | .969 |

FIGURE 1. Dimensions and configurations - Continued.



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TABLE I. Operating characteristics.

| Part no. MS 27750- | Coil data | | | | | | | | | | | Time - (milliseconds maximum) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Nominal | | | Max | | Max pick-up voltage | | | Hold voltage 1/ | Drop out voltage 1/ | Operate 2/ | Release 3/ | Contact Bounce | | | |
| | | | | | | | | | | | | | | Main | | Aux | |
| | | Volts 1/ | Freq Hz | Ω Res Min | Volts | Amp | Normal 1/ | High temp test | Cont current test | | | | | NO | NC | NO | NC |
| 1 | X1,X2 | 28 | dc | 150 | 29 | .190 | 18 | 20 | 21 | 7.0 | 1.5 | 35 | 25 | 3 | --- | --- | --- |
| 2 | X1,X2 | 28 | dc | 150 | 29 | .190 | 18 | 20 | 21 | 7.0 | 1.5 | 35 | 25 | 3 | --- | --- | --- |

1/ Over temperature range.
2/ With nominal coil voltage.
3/ From nominal coil voltage.

TABLE II. Rated contact load (amperes per pole) case grounded.

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, phase | | | | 115/200 V ac, 3 phase 2/ | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | | 400 Hz | 60 Hz | |
| Resistive 1/ | 50 | 25 | | | | 50 | | | | 50 | | | | |
| Inductive | 20 | 15 | | | | | | | | | | | | |
| Inductive | 50 | | | | | 50 | | | | 50 | | | | |
| Motor | 40 | 15 | | | | 30 | | | | 30 | | | | |
| Lamp | 50 | 10 | | | | 15 | | | | 15 | | | | |
| Transfer load | 10 | | | | | 12.5 | | | | 12.5 | | | | 3/ |
| Mechanical life reduced current | 200 | 7 | | | | 14 | | | | 14 | | | | |
| Mixed loads | 50 | 1 | | | | 5 | | | | 5 | | | | |

1/ With nominal coil voltage.
2/ Absence of value indicates relay is not rated for 3 phase applications.
3/ Transfer load indicates relay is suitable for transfer between unsychronized ac power supplies at rating indicated.

Environmental characteristics:

| | |
|---|---|
| Temperature range | -55°C to +71°C |
| Max altitude rating | 50,000 ft |
| Shock g-level | 50 g |
| Duration | 6 ms |
| Max duration contact opening | 10 μs |
| Vibration sinusoidal | |
| G-level | 10 g |
| Frequency range | 70-2,000 Hz |
| Acceleration: | 15 g |



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Electrical characteristics:

| | |
|---|---|
| Insulation resistance, initial | 100 megohms |
| After life or environmental tests | 50 megohms |

Dielectric strength (sea level):

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,000 | 1,000 |
| Aux contacts | N/A | N/A |
| All other points | 1,000 | 1,150 |

Dielectric strength (altitude):

| | 50,000 ft |
|---|---|
| Coil to case | 500 V rms |
| Aux contacts | N/A |
| All other points | 700 V rms |

| | |
|---|---|
| Max contact drop initial | 150 volts |
| After life test | 175 volts |
| Overload current | 100 A dc; 115/200 V ac 400 Hz 400 A |
| Rupture current | 125 A dc; 115/200 V ac 400 Hz 400 A |
| Duty rating | Continuous |

Qualification by similarity: See MIL-PRF-6106.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

APPLICABLE PUBLICATIONS:

Referenced document. This document references MIL-PRF-6106.



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CONCLUDING MATERIAL:

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2026-043)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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